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Factors affecting the infiltration capacity in bamboo groves

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Abstract To determine the effects of the characteristics of litter and surface soil on the infiltration capacity in bamboo groves, infiltration rates were measured on a plot-scale using an oscillating-nozzle rainfall simulator at seven Moso bamboo (*Phyllostachys pubescens*) sites and one Madake (*P. bambusoides*) site in Japan. The maximum final infiltration rate (FIR_{max}), which was defined in this study as the infiltration capacity, was calculated, and the correlation between FIR_{max} and earth surface properties was analyzed. The FIR_{max} ranged from 9 to 308 mm h⁻¹, showing little correlation with litter as surface cover, and a weak correlation with bulk density (0.3–0.9 g cm⁻³) and organic matter content (10.4–45.9%). These results suggest that the bulk density and organic matter content of surface soil, rather than surface cover, can be the controlling factors of infiltration capacity in bamboo groves. The bulk density of surface soil correlated significantly with the soil organic matter content, represented by bamboo rhizomes, suggesting that the infiltration capacity can increase with the amount of root networks present, related to macropore formation. The stand density showed a weak positive correlation, although not significant, with the soil organic matter content and a positive correlation with FIR_{max} . Thus, the infiltration capacity of bamboo groves can increase as the stand density of culms and bamboo rhizomes increase.

Keywords Bamboo grove · Infiltration capacity · Maximum final infiltration rate · Root system · Stand density

Introduction

Bamboo was originally introduced from China to Japan, where bamboo groves were developed and managed for timber and food production. Recently, such bamboo groves have been abandoned owing to the import of bamboo sprouts (called *takenoko* in Japanese) from other countries (Torii 2003) and they have rapidly expanded into and invaded surrounding areas (Torii and Isagi 1997; Nishikawa et al. 2005). These expanding bamboo groves are cause for concern as they may increase the likelihood of landslides and create ecological imbalances (Torii 2003; Yamamoto et al. 2004). Bamboos grow considerably faster than tree species (Ueda 1960), and bamboo roots can easily invade surrounding areas (Kawai et al. 2010). Thus, they dominate both the tree canopy and the surface soil by their root networks, resulting in the decline of other species (Katanoda 2004; Yamamoto et al. 2004; Yokoo et al. 2005; Suzuki et al. 2006). Although clear-cutting may seem to be a best management practice for preventing bamboo expansion, bamboo shoots must be pruned repeatedly to effectively paralyze a bamboo grove, as bamboo rhizomes remain alive after cutting and continuously sprout new shoots (Ueda 1960; Katanoda and Ide 2005; Katanoda 2007). Bamboo grove expansion can convert land cover, which may in turn alter water cycles in forested watersheds (Onda et al. 2010).

Infiltration plays an essential role in the water cycle in forested watersheds (e.g., Nakano 1976; National Research Council 2008). In general, infiltration in a forested surface is affected by many factors, such as surface cover and the

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physical properties of surface soil (e.g., Murai and Iwasaki 1975; Nakano 1976; Yukawa and Onda 1995; Kato et al. 2008; Miyata et al. 2009; Hiraoka et al. 2010). Murai and Iwasaki (1975) examined the factors controlling the infiltration rate in conifer plantations, broadleaf forests, mixed forests, and grassland in Japan. They suggested that the surface cover of litter and surface soil characteristics such as root networks and macropores could be controlling factors. Hiraoka et al. (2010) conducted rainfall simulation experiments at a Japanese cypress plantation and revealed that the infiltration rate could increase with surface cover, which can protect soil from the raindrop impact of throughfall. On a bare surface, raindrop impact may lead to soil sealing or crusting, which can decrease the infiltration rate (Murai and Iwasaki 1975; Yukawa and Onda 1995; Kato et al. 2008; Miyata et al. 2009). Examining the infiltration rate is important in determining the factors controlling infiltration capacity, and the characteristics of surface cover and surface soil, which influence infiltration, differ depending on the tree or bamboo species present.

In bamboo groves, neither the infiltration rate nor its controlling factors have been examined, although recent studies have measured throughfall, stemflow, and soil moisture content (Sakai et al. 1996, 2007; Wang and Liu 2006; Onozawa et al. 2009; Shinohara et al. 2010). Estimation of the infiltration capacity of bamboo groves could contribute to evaluation of the effects of bamboo grove expansion on hydrological cycles. More specifically, rainfall-infiltration process in bamboo groves could be incorporated into runoff models, allowing for assessment of the effects of bamboo grove expansion on water resources in forested watershed areas.

The objectives of this study were to determine the infiltration capacity and to elucidate the factors that control the infiltration capacity in bamboo groves. Under canopies, the raindrop impact of throughfall is greater than that of precipitation (Tsukamoto 1966; Nanko et al. 2004) and can alter the surface soil condition (Yukawa and Onda 1995). Therefore, an infiltration rate measurement method that considers raindrop impact is required. To measure infiltration rates, a rainfall simulation experiment was conducted using an oscillating-nozzle rainfall simulator at eight bamboo sites. We analyzed the correlation between infiltration capacity and earth surface properties to examine the effects of bamboo floor characteristics on infiltration capacity.

Materials and methods

Study area

The study area was located in Yamaguchi Prefecture (6110 km²), western Japan, which ranges from 33°83′ to

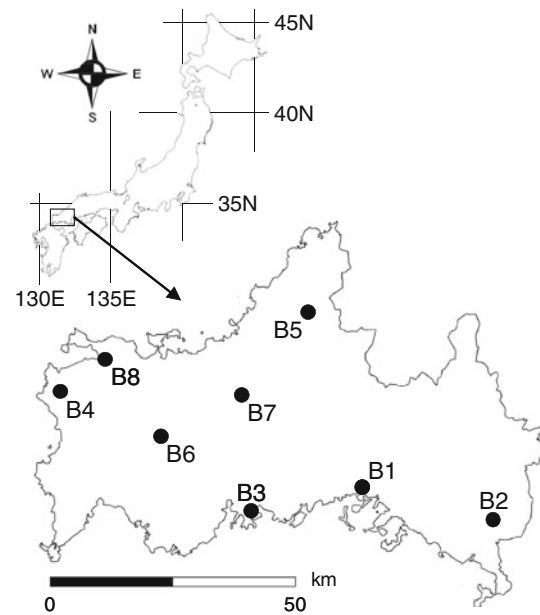


Fig. 1 Location of the study area and studied sites

34°68′N and from 130°86′ to 132°25′E (Fig. 1). The average temperature, mean annual precipitation, and average maximal rainfall intensity from 1968 to 2008 were 15.1°C, 1,863 mm, and 45.4 mm h⁻¹, respectively, at the Yamaguchi meteorological station (34°16′N, 131°45′E) located in the approximate center of Yamaguchi Prefecture. The primary rainy seasons are the Baiu season from June to mid-July and the typhoon season from mid-September to October. Geological features of the area include sedimentary rock, volcanic stone, and pyroclastic rock. Soils are brown forest soils according to the Japanese soil classification system (Cambisols by the US Soil Taxonomy) (Table 1).

Eight bamboo grove sites were selected in Yamaguchi Prefecture and labeled B1–B8 (Fig. 1). The altitude and slope gradient of the selected sites ranged from 30 to 360 m and from 22 to 41°, respectively. The sites were covered by Moso bamboo (*Phyllostachys pubescens*) except for site B8, which was covered by Madake (*P. bambusoides*). Stand age at the time of measurement was 1 year at sites B1–B5 and 2 years at sites B6–B8. Stand density of culms, diameter at breast height of culms, and basal area of culms ranged from 3,004 to 16,299 culms ha⁻¹, from 6.2 to 12.0 m, and from 31.7 to 120.1 m² ha⁻¹, respectively. The relative light intensity, which is the proportion of light intensity of an inside grove to that of an outside grove, was approximately 1% (Table 1). Minimal understory vegetation was observed in the groves, and the A₀-horizon was composed of mostly litter. In addition, a considerable quantity of roots had developed in the 0–4 cm layer of the A-horizon.

Table 1 Site description of studied plots

Geology	Soil type ^a	Direction	Altitude (m)	Slope gradient (°)	Bamboo species	Age (years)	SD (culms ha ⁻¹)	DBH (cm)	BA (m ² ha ⁻¹)	Height (m)	RLI (%)
B1 Triassic	Pelite schist	SE	60	41	Moso bamboo	1	3,004	11.5	31.7	16.5	0.4
B2 Cretaceous	Granite	N	80	34	Moso bamboo	1	4,506	13.5	65.5	14.9	1.6
B3 Cretaceous	Granite	SE	30	22	Moso bamboo	1	7,010	9.7	52.8	13.0	0.1
B4 Tertiary	Sedimentary rock	N	50	29	Moso bamboo	1	6,536	11.5	70.1	13.6	2.0
B5 Permian	Sedimentary rock	NNW	370	40	Moso bamboo	1	8,505	10.5	75.7	11.9	0.7
B6 Cretaceous	Granite	SE	80	34	Moso bamboo	2	10,502	12.0	120.1	13.9	0.5
B7 Cretaceous	Sedimentary rock	SE	60	39	Moso bamboo	2	8,020	11.7	87.4	12.9	1.3
B8 Cretaceous	Granite	NE	40	24	Madake	2	16,299	6.2	50.1	9.6	0.9

SD stand density of culms, DBH mean diameter at breast height of culms, BA basal area of culms, RLI relative light intensity

^a From Forest soil Division (1976)

Rainfall simulation experiments

For the rainfall simulation experiments, an oscillating-nozzle rainfall simulator, which was modified based on the design used by Kato et al. (2008), was used. The rainfall simulator can produce high intensity rainfall, which can generate overland flow throughout a plot. Each plot (1 m² in projected area) was installed in the upper part of the hillslope; the plots did not include holes created by picking bamboo sprouts. The simulator was set up at the plot with the nozzle unit 2.2 m high and vertical to the plot surface. The duration of the experiment was longer than 15 min in order to achieve a steady-state surface runoff rate. Runoff was collected using an aluminum tray at the bottom of the plot and was measured manually every minute, using a volumetric cylinder. The rainfall intensity applied in the experiment was measured at each plot after the experiment by covering the plot with a vinyl sheet and trapping all water sprinkled onto the plot. Trapped water was measured and converted to rainfall intensity based on flow volume. The rainfall intensity of the experiments ranged from 93 to 225 mm h⁻¹ (Table 2). The rainfall intensity at site B4 was less than that at other sites due to simulator trouble.

Calculation of maximum final infiltration rate

Infiltration rate, which is calculated by subtracting the runoff rate from the applied rainfall rate, can gradually approach a steady value with time under a constant rainfall rate, and the steady value is defined as the final infiltration rate (Nakano 1976). The final infiltration rate on a plot-scale in a rainfall simulation experiment generally increases with increasing applied rainfall intensity and gradually approaches a steady-state rate because of the wide range of soils, vegetation composition, and ecological status of a ground surface (Hirata 1956; Murai and Iwasaki 1975; Hawkins 1982; Dunne et al. 1991; Paige and Stone 2003; Stone and Paige 2003). Varying ground surface conditions can result in spatial variability in infiltration capacities on a point-scale across a plot (Hirata 1956; Murai and Iwasaki 1975; Hawkins 1982; Dunne et al. 1991; Paige and Stone 2003). If low intensity rainfall is applied, surface runoff will partially occur within the plot, i.e., only on the surface with lower infiltration capacity. As a result, the final infiltration rate can be lower than the infiltration capacity. Because rainfall simulation experiments using a single rainfall intensity may fail to estimate the infiltration capacity, some mathematical relationships between the rainfall intensity and final infiltration rate have been proposed (Hirata 1956; Hawkins 1982; Tanaka and Tokioka 2007) to calculate infiltration capacity.

Hawkins (1982) formalized a relationship between the measured infiltration rate and application rate, assuming an

Table 2 Rainfall intensity, maximum final infiltration rate, and bamboo floor characteristics

	R (mm h ⁻¹)	$FIR_{\max}$ (mm h ⁻¹)	Litter cover percentage (%)	Dry weight of litter (g m ⁻²)	A_0 thickness (cm)	Bulk density (g cm ⁻³)	SOMC (%)	Particle size distribution (%)			
								Clay	Silt	Sand	Gravel
B1	192	43	95	659	3.0	0.9	10.4	10	39	17	35
B2	199	9	85	511	3.0	0.7	13.3	4	29	30	38
B3	201	187	96	526	5.0	0.6	22.3	8	23	55	15
B4	93	70	90	894	2.0	0.6	18.9	29	36	33	2
B5	200	158	85	704	3.0	0.3	45.9	22	50	21	7
B6	225	308	100	913	5.0	0.4	35.4	9	21	53	17
B7	187	175	90	432	1.5	0.5	30.6	6	16	19	59
B8	187	252	100	571	2.0	0.5	27.2	32	51	14	2

SOMC soil organic matter content, R rainfall intensity, $FIR_{\max}$ maximum final infiltration rate

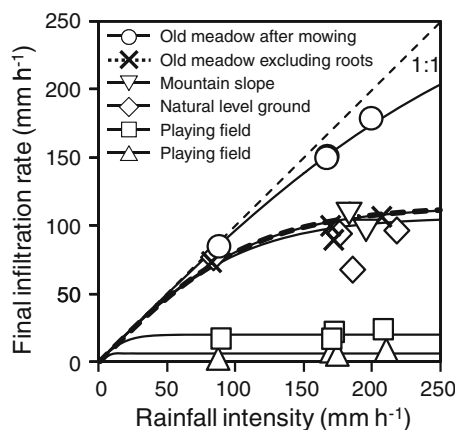


Fig. 2 Observed variation in final infiltration rate with rainfall intensity, as measured by Tanaka and Tokioka (2007). Curves represent the measured data applied to Eq. 1 using the least-squares method

exponential distribution of infiltration capacity over an area. Tanaka and Tokioka (2007) measured final infiltration rates using an oscillating-nozzle rainfall simulator at 22 sites with various land uses in Japan and proposed the following empirical equation based on Hawkins' relationship:

$$FIR_R = FIR_{\max} \tanh\left(\frac{R}{FIR_{\max}}\right) \quad (1)$$

where $FIR_{\max}$ is the maximum final infiltration rate under infinite rainfall intensity, FIR_R is the final infiltration rate under rainfall with an intensity of R , and R is the applied intensity. They examined the relationship between the applied rainfall intensity and final infiltration rate in the rainfall simulation experiments at various land-use sites (Fig. 2). The curves in Fig. 2 represent the data for each site as applied to Eq. 1 using the least-squares method. On the basis of these results, Tanaka and Tokioka (2007) suggested that Eq. 1 can be applied to any kind of land use.

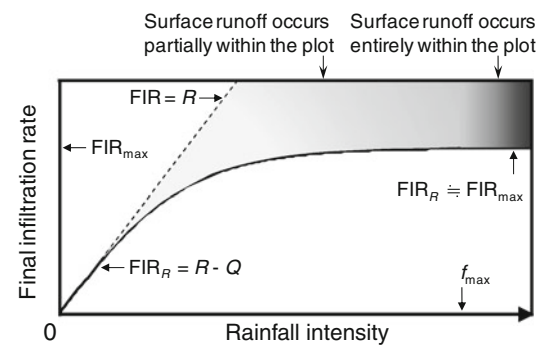


Fig. 3 Graphical representation of the mathematical relationship between rainfall intensity (R) and final infiltration rate (FIR_R) calculated from Eq. 1. $f_{\max}$ represents the maximum infiltration capacity on a point-scale within a target plot (Hawkins 1982)

A graphical representation of Eq. 1 is shown in Fig. 3. A difference between the rainfall intensity and FIR_R indicates the occurrence of surface runoff (shaded area in Fig. 3). For any applied intensity less than $f_{\max}$ (Hawkins 1982), which indicates maximal infiltration capacity on a point-scale, surface runoff partially occurs within the target area. As a result, FIR_R increases with increasing rainfall intensity. Similarly, for any applied intensity more than $f_{\max}$, surface runoff can occur entirely within the target area and the final infiltration rate gradually approaches a steady state, i.e., $FIR_{\max}$, with increased rainfall intensity. Infiltration capacity is defined as the final infiltration rate when rainfall excess occurs entirely within an area (Horton 1945); thus, $FIR_{\max}$ can be regarded as the infiltration capacity.

Because $FIR_{\max}$ allows us to compare the infiltration capacity among the plots, $FIR_{\max}$ was estimated through the numerical analysis in Eq. 1 using the Newton–Raphson method (Kress 1998) for each experiment. The FIR_R was calculated by averaging the infiltration rate over a 5-min period at the end of the experiment.

Sampling

Litter percentage indicates the percentage of the forest floor that is covered with litter in an area within the plots. The litter percentage was measured by image analysis of photographs, using a box counting method (Yang and Juo 2001), in which the photograph was divided into 100 equal parts and the covered areas were counted, to estimate the cover effect on surface mineral soils. Litter materials were collected in an area within each plot, dried for 12 h at 105°C, and then weighed to obtain dry weight values. The thickness of the A_0 -horizon was averaged using measurement values from the A_0 -horizon at several points within each plot, to estimate the development of an organic layer.

To determine the physical properties of surface soil, undisturbed soil core samples were collected at the A -horizon surface within the plots using 400-cm³ steel cylinders. In the laboratory, the soil cores were dried, weighed to measure bulk density, and sieved through a 2-mm mesh sieve. The samples were sieved into size classes of 16.0, 8.0, 4.0, 2.0, 0.85, and 0.425 mm to determine the particle size distribution. The distribution of soil particles less than 0.425 mm in size was obtained using a laser diffraction particle size analyzer (SALD-3100; Shimadzu, Japan). Organic matter in the residue on the 2-mm sieve was weighed as the coarse organic matter content. Subsamples less than 2 mm in size were then ashed for 4 h at 450°C to estimate the fine organic matter content. Soil organic matter content was calculated as the sum of coarse and fine organic matter content and was defined as the total carbon of surface soil.

Analysis

Correlation analysis was conducted between FIR_{max} and earth surface properties to determine the effects of floor conditions on infiltration capacity in bamboo groves. The percentage of litter cover, dry weight of litter, thickness of the A_0 -horizon, and surface soil physical properties at 4-cm depth were used as factors affecting FIR_{max} . The bulk density, organic matter content, and particle size distribution were used as physical properties of the surface soil. Statistical analyses were conducted using SPSS 17.0 software (SPSS Japan).

Results

Infiltration rates and maximum final infiltration rates

The temporal variation in the infiltration rate for each rainfall simulation experiment is shown in Fig. 4. The infiltration rate decreased rapidly for up to 3 min from the

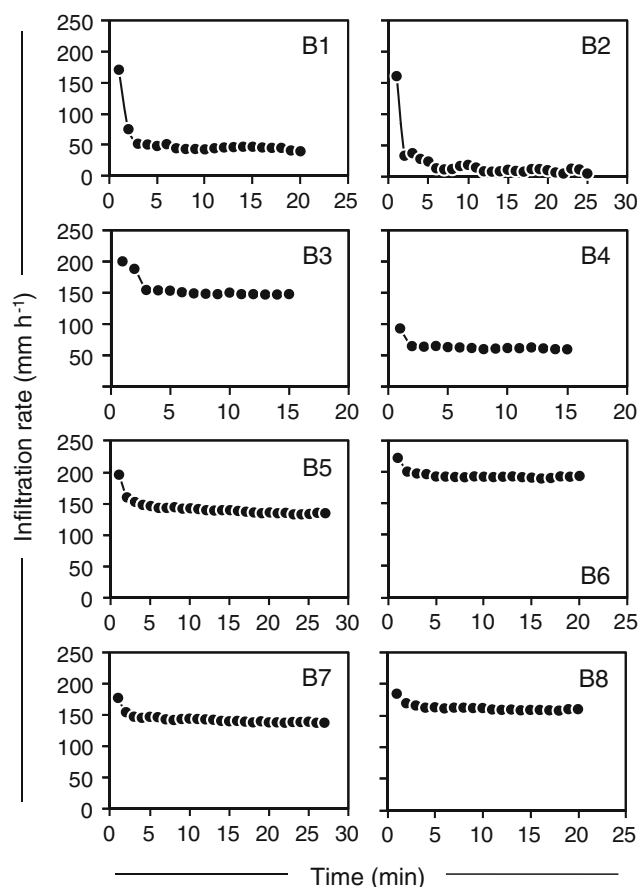


Fig. 4 Infiltration curves of studied plots

start of rainfall and leveled off thereafter, leading to the final infiltration rate at the end of the experiment. In the 1 year stands, although the infiltration rate decreased rapidly at 2 min from the beginning of the experiments in sites B1 and B2, little decline was observed in the infiltration curves for sites B3, B4, and B5. In 2-year stands, the decline in the infiltration rate was less than that in 1-year stands (Fig. 4). The FIR_{max} estimated by Eq. 1 ranged from 9 to 308 mm h⁻¹ (Table 2).

Bamboo floor characteristics

In all the study sites, there was minimal bare area on the bamboo floor, which was covered with bamboo litter, resulting in a high litter cover percentage of over 85%. The dry weight of litter ranged from 432 to 913 g m⁻². The A_0 -horizon was well developed, and its thickness at each plot ranged from 1.5 to 5.0 cm (Table 2).

The bulk density and organic matter content of surface soil (4 cm depth under the A_0 -horizon) ranged from 0.3 to 0.9 g cm⁻³ and from 10 to 46%, respectively (Table 2). The bulk density seemed to be lower than that of surface soil in forested areas of Japan, at approximately

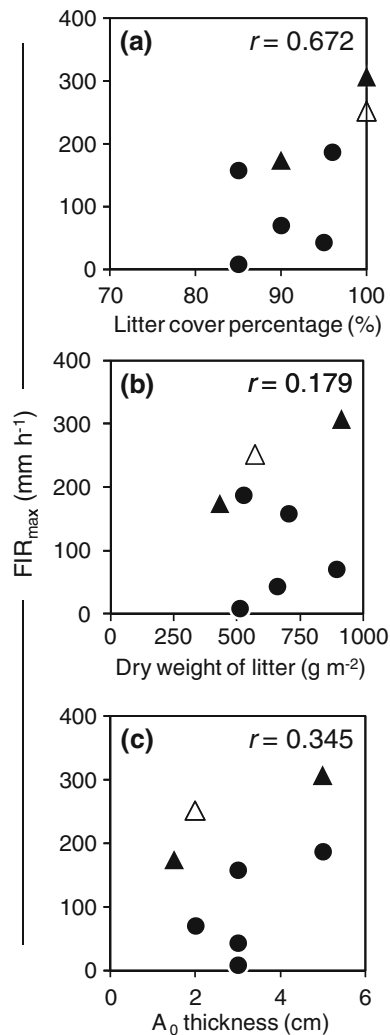


Fig. 5 Relationships between litter and maximum final infiltration rate (FIR_{max}). **a** Litter cover percentage, **b** dry weight of litter, and **c** A_0 thickness. Solid circle 1-year stand of Moso bamboo, solid triangle 2-year stand of Moso bamboo, open triangle 2-year stand of Madake

1.0 g cm⁻³ (Matsunaka 2003). Significantly lower bulk density of surface soil in bamboo groves may be attributed to the high organic matter content or soil porosity. The soil particle size distribution varied among the sites (Table 2). Even though the particle size distributions were similar between sites B3 and B6 and between sites B5 and B8, the bulk densities varied, suggestive of the difference in organic matter content.

Relationships between maximum final infiltration rate and bamboo floor characteristics

To examine the relationship between FIR_{max} and litter conditions, correlation analyses were conducted between FIR_{max} and the litter cover percentage, dry weight of litter, and thickness of the A_0 -horizon. No significant correlation

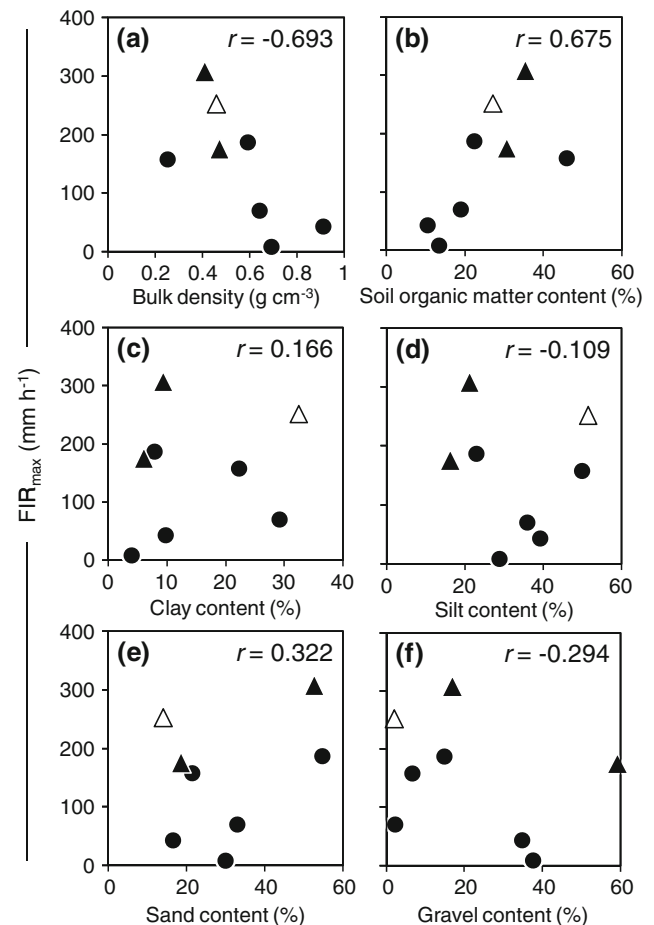


Fig. 6 Relationships between physical properties of surface soil and maximum final infiltration rate. **a** Bulk density, **b** soil organic matter content, **c** clay content, **d** silt content, **e** sand content, and **f** gravel content. Solid circle 1-year stand of Moso bamboo, solid triangle 2-year stand of Moso bamboo, open triangle 2-year stand of Madake

of FIR_{max} was found with any of these factors ($P = 0.07, 0.67, 0.40$) (Fig. 5). Furthermore, few correlations were found between FIR_{max} and each surface cover factor in the 1-year stands. Although the dataset was small for the 2-year stands, positive correlations were found when the measured data were separated by stand age. These results suggest that FIR_{max} in bamboo groves may be affected not only by litter conditions at the soil surface but also by the stand age.

The relationships between FIR_{max} and the physical properties of the surface soil were examined (Fig. 6). The FIR_{max} decreased with bulk density ($P = 0.06$), increased with organic matter content ($P = 0.07$), and showed no correlation with either particle-size fraction ($P > 0.44$). No significant correlations were found when the measured data were separated by stand age. These results suggest that FIR_{max} can be related to bulk density and organic matter content, but not to particle size, in bamboo groves.

Discussion

The wide variation in FIR_{max} found in this study suggests heterogeneity of the slope surfaces of bamboo groves. To compare FIR_{max} values between this study and previous studies, FIR_{max} was calculated from the final infiltration rates reported in the literature using Eq. 1; the calculated values are listed in Table 3. From Murai and Iwasaki (1975), who measured infiltration rates using a mist-type rainfall simulator, FIR_{max} was calculated to be 108 mm h^{-1} for bamboo grass (*Sasamorpha borealis*), 151 mm h^{-1} for larch plantation (*Larix leptolepis*), 316 mm h^{-1} for broad-leaf forest (*Quercus serrata*, *Padus grayana*), and 486 mm h^{-1} for turf (*Zoysia japonica*). In grasslands, FIR_{max} was found to range from 56 to 104 mm h^{-1} in experiments using an oscillating-nozzle rainfall simulator in semi-arid areas in Australia (Loch 2000) and Mongolia (Kato et al. 2009). Thus, the infiltration rate can vary depending on the vegetation covering the slope surface. However, the infiltration rate can also vary in areas of similar vegetation and cover conditions. Hiraoka et al. (2010) measured the infiltration rates in seven plots covered 80–100% with ferns (*Gleichenia japonica*) in a Japanese cypress plantation. They reported a wide range of FIR_{max} , from 92 to 322 mm h^{-1} , and suggested that root networks may affect the variation in infiltration rates. It is widely recognized that fine and decaying roots in root networks create macropores in the soil, and that macropores in surface soil can contribute to infiltration (Karizumi 1979). The variability of FIR_{max} shown in the present study was similar to that reported by Hiraoka et al. (2010), suggesting that the root networks causing macropores can be a controlling factor of infiltration rates.

Although not significant, but showing a weak tendency, FIR_{max} exhibited correlations with the bulk density and organic matter content of the surface soil, suggesting that the surface soil characteristics may affect the infiltration capacity. The bulk density of surface soil showed a highly significant negative correlation with organic matter content ($P < 0.01$), indicating that bulk density is strongly affected by the organic matter content of the surface soil (Fig. 7). Observations of soil core samples revealed a large amount of coarse organic matter, most of which was composed of bamboo rhizomes. Therefore, the high organic matter content of the surface soil can be attributed to bamboo rhizomes. The amount of bamboo roots in the surface soil can thus be an important factor controlling infiltration capacity.

In general, root systems can enhance soil aggregation, resulting in the formation of intra-aggregate pores in the surface soil (Karizumi 1979; Kobayashi 1980; Beven and Germann 1982). These pores are known to be used by preferential flow (Beven and Germann 1981, 1982; Noguchi et al. 1997; Ghodrati et al. 1999; Noguchi 2001),

which can enhance infiltration. The high distribution density of bamboo rhizomes found in the present study may form intra-aggregate pores, resulting in an acceleration of infiltration capacity.

A higher infiltration capacity was found in older bamboo stands (Figs. 5 and 6), suggesting that bamboo stand age may determine the density of roots in surface soil. Bamboo rhizomes grow and branch annually (Kawai et al. 2008), sprouting approximately $2,500 \text{ shoots ha}^{-1} \text{ year}^{-1}$ even after clear-cutting (Katanoda and Ide 2005; Katanoda 2007). This vigorous growth of bamboo indicates that the stand and rhizome density of bamboo can increase annually, resulting in higher stand and rhizome density in older stands. In Fig. 8a, an outlier can be seen from the 2-year stand of Madake (site B8). Among 2-year stands, site B8 had the smallest basal area of culms, despite having the highest stand density of culms (Table 1). The basal area of culms showed a direct proportional relationship with belowground biomass (Kiyono et al. 2007), suggesting that the organic matter content of site B8 was less than that of sites B6 and B7. Considering only the Moso bamboo data, significant correlation was found between the stand density and the organic matter content ($r = 0.86$, $P = 0.01$, $n = 7$), suggesting that the organic matter content of the Moso bamboo site can be estimated from the stand density of culms. The FIR_{max} showed a positive correlation with the stand density ($P < 0.05$) (Fig. 8b) and, in the case of the Moso bamboo data, a strong correlation was observed ($r = 0.90$, $P < 0.01$, $n = 7$). These results indicate that the increase in bamboo rhizomes may contribute to higher organic matter and lower bulk density, which are related to macropore formation, as the bamboo culms become denser, thereby producing a higher infiltration capacity.

Conclusions

To determine the infiltration traits of expanding bamboo groves, the infiltration rate was measured using an oscillating-nozzle rainfall simulator at eight experimental plots on bamboo hillslopes, and correlation analysis between infiltration capacity and earth surface properties was conducted. In bamboo groves with high litter cover and a well-developed A_0 -horizon, the maximum final infiltration rate varied widely and showed weak correlation with the bulk density and organic matter content of surface soil, suggesting that bulk density and organic matter can be factors controlling infiltration capacity. The macro-organic matter in the surface soil was composed mainly of bamboo rhizomes, indicating that the amount of bamboo root networks can also influence infiltration capacity. The weak correlation between organic matter content of surface soil and stand density of culms suggests that the infiltration capacity

Table 3 Results of field rainfall simulation experiments of the current and previous studies

Climate	Land cover	Surface cover	Surface cover percentage (%)	Rainfall simulator	R (mm h ⁻¹)	$FIR_{\max}$ (mm h ⁻¹)	References
Bamboo grove							
Humid temperate	<i>Phyllostachys pubescens</i>	Bamboo leaf	95	Oscillating nozzle type	192	43	This study
Humid temperate	<i>Phyllostachys pubescens</i>	Bamboo leaf	85	Oscillating nozzle type	199	9	This study
Humid temperate	<i>Phyllostachys pubescens</i>	Bamboo leaf	96	Oscillating nozzle type	201	187	This study
Humid temperate	<i>Phyllostachys pubescens</i>	Bamboo leaf	90	Oscillating nozzle type	93	70	This study
Humid temperate	<i>Phyllostachys pubescens</i>	Bamboo leaf	85	Oscillating nozzle type	200	158	This study
Humid temperate	<i>Phyllostachys pubescens</i>	Bamboo leaf	100	Oscillating nozzle type	225	308	This study
Humid temperate	<i>Phyllostachys pubescens</i>	Bamboo leaf	90	Oscillating nozzle type	187	175	This study
Humid temperate	<i>Phyllostachys bambusoides</i>	Bamboo leaf	100	Oscillating nozzle type	187	252	This study
Natural forest							
Humid temperate	<i>Quercus serrata</i> and <i>Prunus grayana</i>	–	–	Mist type	180	316	Murai and Iwasaki (1975)
Plantation forest							
Humid temperate	<i>Larix leptolepis</i>	–	–	Mist type	180	151	Murai and Iwasaki (1975)
Humid temperate	<i>Chamaecyparis obtusa</i>	<i>Gleichenia japonica</i>	80	Oscillating nozzle type	169	93	Hiraoka et al. (2010)
Humid temperate	<i>Chamaecyparis obtusa</i>	<i>Gleichenia japonica</i>	84	Oscillating nozzle type	169	124	Hiraoka et al. (2010)
Humid temperate	<i>Chamaecyparis obtusa</i>	<i>Gleichenia japonica</i>	88	Oscillating nozzle type	181	162	Hiraoka et al. (2010)
Humid temperate	<i>Chamaecyparis obtusa</i>	<i>Gleichenia japonica</i>	98	Oscillating nozzle type	168	92	Hiraoka et al. (2010)
Humid temperate	<i>Chamaecyparis obtusa</i>	<i>Gleichenia japonica</i>	100	Oscillating nozzle type	169	101	Hiraoka et al. (2010)
Humid temperate	<i>Chamaecyparis obtusa</i>	<i>Gleichenia japonica</i>	100	Oscillating nozzle type	204	253	Hiraoka et al. (2010)
Humid temperate	<i>Chamaecyparis obtusa</i>	<i>Gleichenia japonica</i>	100	Oscillating nozzle type	340	322	Hiraoka et al. (2010)
Grass land							
Humid temperate	<i>Sasamorpha purpurascens</i>		–	Mist type	180	108	Murai and Iwasaki (1975)
Humid temperate	<i>Zoysia japonica</i>		–	Mist type	180	486	Murai and Iwasaki (1975)
Semiarid	<i>Chloris gayana</i> and <i>Pennisetum clandestinum</i>		100	Oscillating nozzle type	130	56	Loch (2000)
Semiarid	<i>Chloris gayana</i> and <i>Pennisetum clandestinum</i>		100	Oscillating nozzle type	130	104	Loch (2000)
Semiarid	<i>Leymus chinensis</i> , <i>Stipa krylovii</i> , and <i>Kochia prostrata</i>		92	Oscillating nozzle type	180	96	Kato et al. (2009)

– Data could not be found in the paper

 R rainfall intensity, $FIR_{\max}$ maximum final infiltration rate

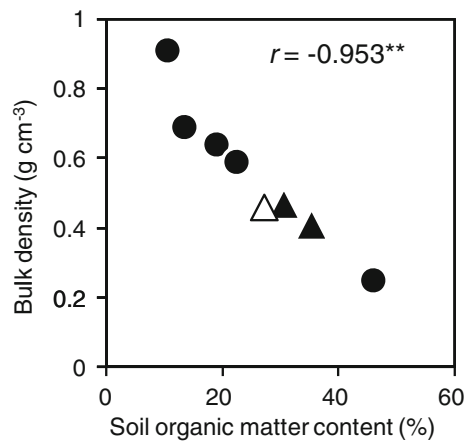


Fig. 7 Relationship between soil organic matter content and bulk density. *Solid circle* 1-year stand of Moso bamboo, *solid triangle* 2-year stand of Moso bamboo, *open triangle* 2-year stand of Madake (** $P < 0.01$)

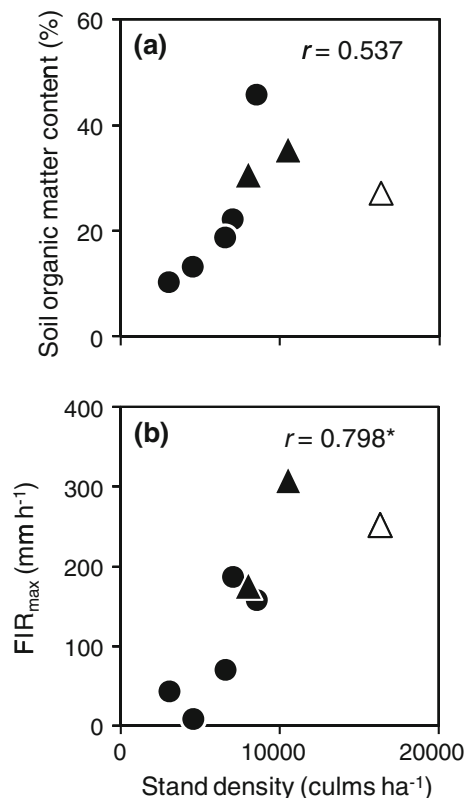


Fig. 8 Relationships between stand density and **a** soil organic matter content and **b** maximum final infiltration rate (FIR_{max}). *Solid circle* 1-year stand of Moso bamboo, *solid triangle* 2-year stand of Moso bamboo, *open triangle* 2-year stand of Madake (* $P < 0.05$)

can increase with bamboo grove age through the increase in stand density of bamboo culms and/or rhizomes. Thus, the stand density of culms was found to be an appropriate index for estimating infiltration capacity. In this study, the infiltration capacity was evaluated on a plot-scale on a

bamboo floor with no holes created by picking bamboo sprouts. For a better interpretation of the hydrological processes in bamboo groves, further studies are needed to evaluate the infiltration capacity of older stands, stands of different bamboo species, and broader areas including holes from bamboo harvesting, as well as to determine the flow pathways after infiltration.

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